INCH-POUND
MIL-C-24320A(SH)
11 April 1990
SUPERSEDING
MIL-C-24320(SHIPS)
22 November 1967
(See 6.6)

MILITARY SPECIFICATION

CLOTH, ACETATE, FLUORESCENT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1 1 Scope. This specification establishes the requirements for fluorescent dyed acetate cloth used in the manufacture of flags for dan buoys and master reference buoys.

1.2 Colors The fluorescent acetate cloth shall be furnished in the following colors as specified (see 6.2):

Color R - Daylight fluorescent red
Color Y - Daylight fluorescent yellow

2. APPLICABLE DOCUMENTS

2 1 Government documents

2 1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-1133 - Packaging of Synthetic Fiber Fabrics

MILITARY

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation.

TEXTILE FIBERS PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 25380)

2 2 Non-Government publications The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

Hardy - Handbook of Colorimetry

(Application for copies should be addressed to the Technology Press, Massachusetts Institute of Technology, Cambridge, MA 02139.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents These documents also may be available in or through libraries or other informational services)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4 4

3 2 Material The material used in the manufacture of fluorescent acetate cloth shall be bright non-pigmented acetate, fluorescent dyed so that the finished product shall meet the requirements of this specification.

3 2.1 Recovered materials Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Weave. The weave shall be plain with single picks in the filling and two warp ends weaving as one

3 4 Properties The properties of the finished cloth shall be as specified in table I and in the following paragraphs when tested in accordance with FED-STD-191 (see 4 5.2)

TABLE I. Properties of fluorescent acetate cloth

Characteristic	Requirement
Weight, ounces per square yard	7.3 ± 0.3
Yarns per inch, min	
Warp	100
Filling	38
Breaking strength ravel strip method, pounds, min	
Warp	78
Filling	70
Water resistance, spray rating, min	90, 90, 80
Colorfastness to	
Crocking	Pass
Accelerated weathering	Good
Salt water	Good

3.5 Dimensions.

3 5 1 Width Unless otherwise specified (see 6 2), the minimum width of the cloth, including commercial width selvages, shall be 39 inches.

3.5 2 Length and put up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up on full width rolls and marked as specified in PPP-P-1133

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3.6 Color The cloth shall be piece dyed; it shall be furnished in one of the following listed colors as specified (see 6.2) and the appearance to the eye in daylight or visible white light shall be as listed alongside the color name

Daylight fluorescent red - purplish red
Daylight fluorescent yellow - yellowish orange.

3 6 1 Spectrophotometric properties The color of the daylight fluorescent cloth shall conform to either the spectrophotometric properties or to the alternate spectrophotometric properties specified in table II, as applicable for the color specified (see 6.2), when determined as specified (see 4.5.2).

TABLE II Spectrophotometric properties of daylight fluorescent cloth colors

Wavelength of light reflected and emitted, nanometers	Reflectance expressed as a percentage of light which would reflect from magnesium oxide			
	Red		Yellow	
	min	max	min	max
400	12	25	--------	14
420	--------	--------	--------	12
440	16	30	--------	--------
450	20	35	--------	--------
480	--------	--------	25	35
490	--------	--------	22	35
520	--------	--------	12	24
530	--------	8	10	22
540	--------	8	--------	--------
550	--------	8	--------	--------
560	--------	--------	--------	--------
580	--------	--------	115	--------
590	--------	--------	120	--------
600	--------	--------	115	--------
610	--------	--------	--------	--------
620	--------	--------	90	--------
630	135	--------	--------	--------
640	135	--------	75	--------
650	135	--------	--------	--------
680	105	--------	65	--------

Alternate spectrophotometric properties of daylight fluorescent cloth colors		
Characteristic of light reflected and emitted	Red	Yellow
Dominant wavelength, nanometers	--------	580 to 589
Complement of dominant wavelength, nanometers	490 to 500	--------
Purity, percent	57 to 66	65 to 75
Brightness, percent, minimum	23	60

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3.7 Workmanship. Workmanship shall be first class in every respect

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material

4.1.2 Material certification When specified in the contract or order, a certificate of compliance shall be prepared (see 6.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows·

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Sampling

4.3.1 First article sample. A sample 4 yards in length, full width, shall be submitted or made ready for the contracting officer or his authorized representative for inspection in accordance with 4 4.

4 3 2 Quality conformance inspection sampling

4 3.2 1 Lot For purposes of sampling for examination and tests, a lot shall consist of all cloth made under essentially the same conditions and offered for delivery at one time

4 3 2 2 Sampling for visual and dimensional examination. A random sample of cloth shall be selected from each lot for visual and dimensional examination in accordance with table III Not more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls

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TABLE III. Sampling for visual and dimensional examination.

Lot size in yards	Sample size in rolls 1/
Up to 1300	3
1301 to 3200	5
3201 to 8000	7
8001 to 22000	10
22,001 to 110,000	15
110,001 and over	25

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined

4.3.2.3 Sampling for tests Samples for testing shall be selected in accordance with table IV. The unit of product for testing shall be 3 yards, full width, of the finished cloth. Random sampling shall be accomplished so that, if possible, each 3-yard sample shall be taken from a different roll and tagged to identify it with the roll sampled.

TABLE IV Sampling for tests.

Lot size in yards	Number of samples	Maximum number of defects available in sample
Up to 800	2	0
801 to 22,000	3	0
22,001 and over	5	1

4 4 First article inspection.

4 4.1 Examination. The sample selected in accordance with 4.3.1 shall be subjected to visual examination for appearance, color and finish as specified in 4.5.1.

4.4.2 Testing. The sample selected in accordance with 4.3.1 shall be subjected to the tests specified in 4.5.2.

4 5 Quality conformance inspection

4.5.1 Visual and dimensional examination

4.5.1.1 Yard by yard examination. Thirty yards from each roll in the sample selected in accordance with table III shall be examined on the face side Defects as specified in section III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as specified below. No linear yard (increment of 1 yard on the measuring device of the inspection machine) shall be penalized more than 4 points The lot shall be unacceptable if the points per 100 square yards exceed 25.0. Point computation shall be as follows

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$$\frac{\text{Total points scored in sample size x 3600}}{\text{Contracted width of cloth (inches) x sample size (yards)}} = \text{Points per 100 square yards}$$

Demerit points - Demerit points shall be assigned as follows.

For defects 3 inches or less in any direction - 1 point
For defects exceeding 3 inches, but not exceeding 6 inches in any direction - 2 points
For defects exceeding 6 inches, but not exceeding 9 inches in any direction - 3 points
For defects exceeding 9 inches in any direction - 4 points

The following defects, when present, shall be scored 4 points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Objectionable odor
Width of cloth less than minimum specified
Poor dye penetration, mottled, streaky or cloudy

4 5 1 2 Examination for identification of pre-shrinkage process and compliance with Textile Fiber Products Identification Act During the yard by yard examination, each roll in the sample shall be examined. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or the ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4 5.1.3 Examination for length. During the yard by yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified shall be considered a defect with respect to length The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length. The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of lengths marked on the ticket.

4.5.2 Test procedures. Each sample of cloth selected in accordance with table IV shall be tested with the methods specified in table V.

TABLE V. Methods of testing.

Characteristic	Requirement	Test method FED-STD-191
Weight, ounce per square yard	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5104
Water resistance, spray	Table I	5526
Colorfastness	Table I	5651, 5671, 5632
Weave	3.3	Visual
Spectrophotometric properties	3 6 1 and table II	4 5 2.1 or 4 5 2 2 (herein)

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4.5.2.1 Spectrophotometric properties. Specimens of cloth shall be cut out at a distance of not less than 6 inches in from the selvage and each end of the sample. The specimens shall be cut from samples which have been protected from exposure to light A color analyzer shall be used. The specimen shall be directly illuminated, normal to the surface exposed, by the full spectral emission from a light source having the spectral quality of CIE (Commission Internationale de l'Eclairage or International Commission on Illumination) Source "C" The light reflected and emitted by the fluorescent material shall be picked up with a cylindrical mirror type apparatus as shown on figure 1 and shall be evaluated spectrophotometrically with the instrument using a slit width of 0.1 millimeter. The Beckman DU spectrophotometer, the Beckman DK spectrophotometer, or equivalent may be used as the instrument for obtaining the spectral data. The light shall be measured at wave lengths specified in table II and expressed in terms of a percentage of the light which would be reflected from the reference standard, a freshly prepared magnesium oxide surface, under similar conditions. To minimize the effects of directional reflectance, readings at two positions shall be averaged (the original sample position and a position where the sample has been rotated through 90 degrees). Readings shall be taken on the face side of the cloth specimen backed up by three thicknesses of the cloth sample to minimize differences in cloth texture and background.

4 5 2 2 Alternate spectrophotometric properties As specified in 4 5 2 1, the percent reflectance shall be determined for the fluorescent cloth by measuring light reflected and emitted at each of the following 30 wavelengths: 422, 432, 436, 439, 444, 450, 456, 461, 462, 469, 478, 489, 495, 515, 530, 541, 544, 552, 562, 564, 573, 577, 585, 589, 600, 601, 611, 624, 627, and 646 nanometers. The measurements at 436, 461, 544, 564, 577, 589, 600, 611, 624, and 646 nanometers shall be added together. The sum shall be multiplied by 0.098 The product shall equal "X".

4 5 2.2 1 "Y" brightness. The measurements at 489, 515, 530, 541, 552, 562, 573, 585, 601, and 627 nanometers shall be added together. The sum shall be multiplied by 0 100. The product shall equal the percent brightness "Y" of the reflected and emitted light.

4.5.2.2 2 "Z" brightness The measurements at 422, 432, 439, 444, 450, 456, 462, 469, 478, and 495 nanometers shall be added together. The sum shall be multiplied by 0 118. The product shall equal "Z"

4.5.2.2 3 Dominant wavelength and purity of light. Dominant wavelength and purity of the light reflected and emitted by the daylight fluorescent dyed cloth shall be determined by consulting charts in the Handbook of Colorimetry by Hardy, after values for x and y are calculated with the following equations:

$$x = \frac{X}{X + Y + Z} \qquad y = \frac{Y}{X + Y + Z}$$

In the case of the daylight fluorescent red color, the complement of the dominant wavelength is numerically identical with the dominant wavelength determined by the above method

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4 6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5 1 1 Navy fire-retardant requirements

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use
Level C - Type I - non-weather resistant.
Category 1 - general use

(b) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Cloth (see 3.5).

5.2.1 Put-up, preservation, packing, and marking. Cloth put-up in rolls (see 3.5) shall be put-up level A or commercial; preservation level A or commercial; packing level A, B or commercial; marking including bar coding and other packaging acquisitioning options as specified (see 6.2) in accordance with PPP-P-1133.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use The acetate cloth covered by this specification is intended for use in flags for dan buoys and master reference buoys.

6.2 Acquisition requirements Acquisition documents must specify the following:

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(a) Title, number and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2 1.1 and 2.2).
(c) Color (see 1.2, 3.6, and 3.6.1).
(d) Whether first article inspection is required (see 3.1).
(e) Dimensions and roll put-up if other than as specified (see 3.5)
(f) When fire-retardant requirements are not required (see 5.1.1(a) and (b)).
(g) Level of put-up, level of preservation and packing, and other packaging options required (see 5.2.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should by reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27 475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4 1 2	DI-E-2121	Certificate of Compliance	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6 4 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

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6.5 Subject term (key word) listing

Colorfastness
Spectrophotometric
Weave

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 8305-N257)

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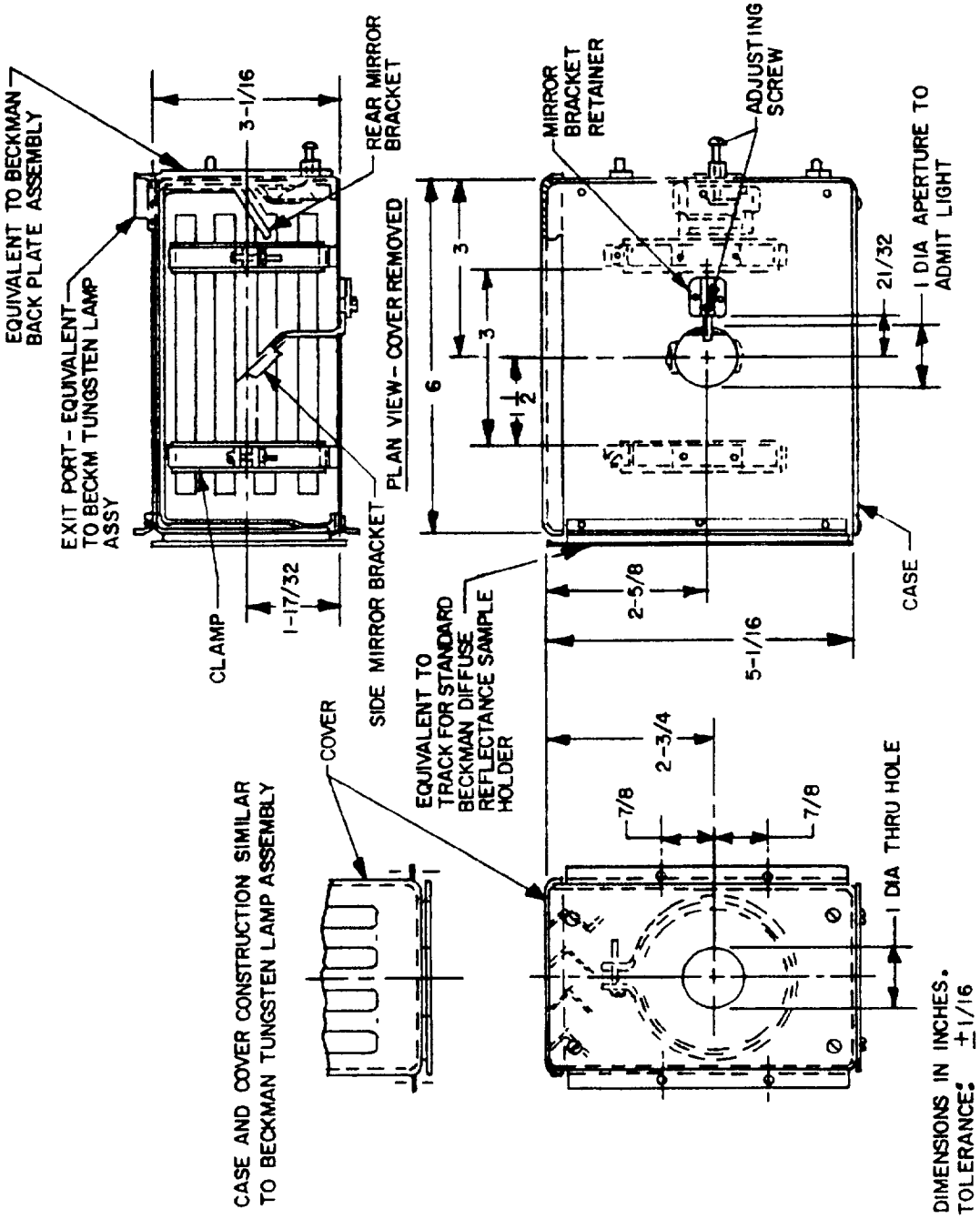


FIGURE 1. Cylindrical mirror apparatus.

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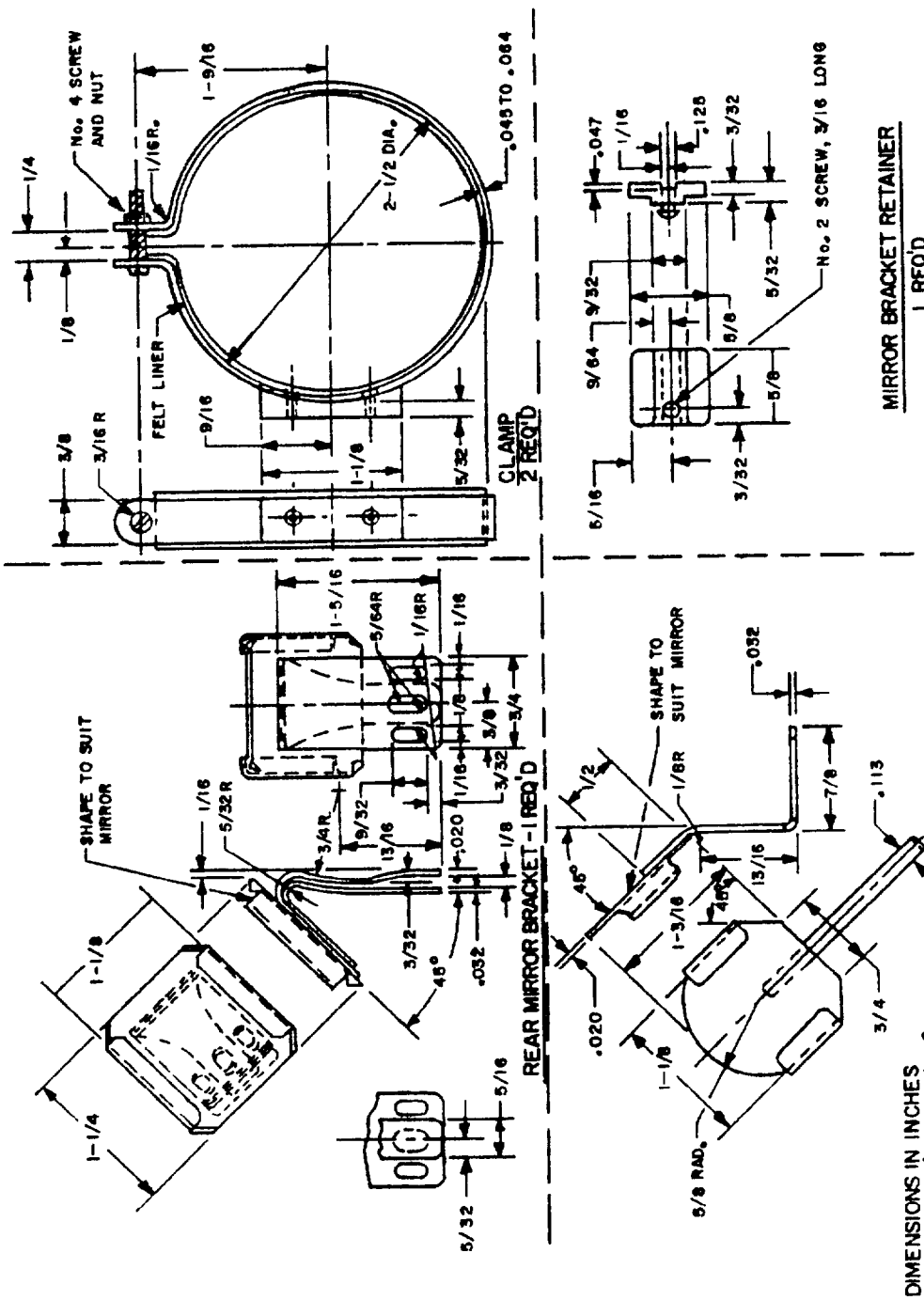


FIGURE 1. Cylindrical mirror apparatus - Continued.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE. This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-24320A(SH)	

3. DOCUMENT TITLE
CLOTH, ACETATE, FLUORESCENT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Reams (PMS 407D34) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 703-602-4717	8-332-4717
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289 2340	

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